

PAPER The MetroVolt Data-Center Burner: Direct-DC Campus Power, Q_E Closure, and the Plug Requirement in a Low-Neutron D– ^3He Tandem Mirror

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Abstract

The load that a hyperscale artificial-intelligence campus imposes on the grid is native direct current, growing faster than transmission can be built, and increasingly sited where interconnection queues are measured in years. We present MetroVolt, a data-center power plant built on the Kronos deuterium–helium-3 (D– ^3He) tandem-mirror burner, whose direct energy converter (DEC) delivers high-voltage direct current *natively*—the same physical form the compute hall consumes—so that the plant couples to the load through two-to-three power-conversion stages rather than the four-to-six of a thermal generator feeding an alternating-current (AC) grid and an uninterruptible supply. At the frozen design point the burner reaches a plant energy multiplication $Q_E = 1.318$ at fusion power 4298.5 MW with a neutron fraction of only 5.44 % of the total power—the low-activation advantage of the fuel—and a length ladder delivers net electric output of +104/ +850/ +2832 MW_e at 55/440/1400 m device length, with a Monte-Carlo uncertainty quantification returning $P(Q_E > 1) = 1.0$ over the design ranges. The binding physics gate is the end-plug potential: confining the central-cell ions requires a plug-to-central-cell density ratio $n_p/n_c = 16$ built from an ion-confining potential $\phi_i = 248.75$ kV against a carrier temperature $T_e = 89.7$ keV, a requirement-class target more than two orders of magnitude beyond the demonstrated tandem-mirror record and retired only by a dedicated plug-potential experiment; direct energy conversion is a bounded upside lever, taken at a deliberately conservative module efficiency $\eta_{\text{DEC}} = 0.49$, not the closure mechanism. Reduced to the data-center product, the direct-DC path raises the source-to-chip efficiency from 75.9 % (conventional AC) to 88.1 % (DEC-direct medium-voltage DC); a plasma-independent electronic buffer follows synchronized GPU training swings at ~ 80 MW s^{−1}—about 105× any plasma throttle—while holding the DC bus to ± 0.40 %; waste-heat-driven absorption cooling takes the facility from PUE 1.35 to 1.17; and the neutron-lean 90 %-aneutronic burn licenses against a mobile tritium source term of ~ 3.6 g, roughly 275× below a deuterium–tritium plant of the same power. We separate throughout what is robust—the neutron-source performance and the power-electronics topology—from what is conditional on retiring the plug gate.

Keywords: tandem mirror; deuterium–helium-3; direct energy conversion; data-center power; HVDC-native distribution; plug potential; low-neutron fusion; behind-the-meter

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Nomenclature

Symbol	Definition
Q_E	plant energy multiplication (gross electric out / recirculating in)
f_n	neutron fraction of total fusion power
n_p/n_c	plug-to-central-cell density ratio (the binding gate)
ϕ_i	ion-confining electrostatic plug potential
T_e	carrier (plug) electron temperature that holds the potential
T_i	central-cell ion temperature
R_m	plug mirror ratio
f_w	warm-ion fill fraction inside the loss cone
η_{DEC}	direct-energy-converter module efficiency (declared, conservative input)
η_{th}	thermal-cycle efficiency on the residual (non-direct) power
P_{fus}, P_n	fusion power; neutron power
P_{ch}	recoverable charged-particle (axial) power
P_{plug}	recirculating power to sustain the plug and thermal barrier
η_{chain}	source-to-chip electrical delivery efficiency
PUE	power-usage effectiveness, $P_{\text{facility}}/P_{\text{IT}}$
CF	plant capacity factor
n^*	synchrotron reabsorption cutoff harmonic

1 Introduction

1.1 *The data-center load is a fusion problem in disguise*

Artificial-intelligence compute has made the data center one of the fastest-growing electrical loads in the industrialized world, and its demand has three features that a conventional generator serves badly. It is *native direct current*: a modern GPU hall converts the incoming alternating current down through an uninterruptible-power-supply (UPS) double conversion and a rack power-supply unit to a low-voltage DC bus at the silicon, discarding several percent of the delivered energy at each stage. It is *transient*: synchronized large-model training draws crest-factor swings of order 1.2 on second timescales that a thermal plant cannot follow with the plasma alone. And it is *sited ahead of the grid*: hyperscale campuses are increasingly built where land and cooling are available rather than where transmission is, so interconnection queues of order four years now gate time-to-power. A generator that produced its output as regulated high-voltage DC, followed a load electronically without touching the plasma, and could sit behind the meter would match this load in a way that no steam turbine feeding an AC grid does.

The Kronos D–³He tandem-mirror burner is exactly such a generator. Its direct energy converter (DEC) recovers charged-particle power *as electricity, directly, without a thermal cycle*, and the natural output of an electrostatic decelerator is high-voltage direct current—the same physical quantity the compute hall wants. This paper is the data-center product realization of that burner, which we call MetroVolt. It shares its plasma physics identically with the Aegis forward-base burner [37]: the two are one machine in two housings, and the physics closure, the binding plug gate, and the low-neutron advantage are common to both. What distinguishes MetroVolt is the coupling to the load—the HVDC-native power-electronics path, the electronic load-following, the waste-heat integration, the power density, and the urban siting case—and that is where this paper concentrates, after establishing the shared physics on a common footing.

1.2 *Why a low-neutron mirror, and what it costs*

A D–³He mirror is attractive because it produces few neutrons and difficult because it needs a large electrostatic plug to confine its ions and a fuel that is scarce on Earth [1, 2, 3]. The low neutron fraction ($f_n = 5.44\%$, from residual D–D side reactions rather than the aneutronic primary channel) is what lets a fusion source sit next to a data-center campus: it minimizes activation, shielding mass, and the mobile tritium inventory that dominates a licensing case [4]. The cost of that advantage is a demanding confinement problem—the tandem-mirror plug—and a fuel-supply problem for ³He that is a separate, lunar-gated matter treated in a companion paper [44] and outside the scope of the present power-plant design. We are explicit throughout about which results are robust and plug-independent (the neutron source, the power-electronics topology) and which are conditional on retiring the plug gate (every net-power number).

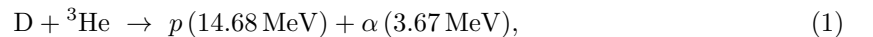
1.3 *Contribution*

This paper contributes: (i) a self-contained statement of the burner’s governing equations and the formal power-balance problem, with the binding plug-potential requirement written as an explicit exponential gate (§2); (ii) the GPU-accelerated computational methodology by which each gate was reduced, naming the real codes (§3); (iii) the direct-DC data-center product physics—the HVDC-native delivery chain, the electronic load-following, the power-usage-effectiveness pathway, the power density, the capacity-factor firming case, and the urban siting and tritium source term—each grounded in a frozen register anchor (§4); (iv) a results section of eleven figures and supporting tables (§5); and (v) a quantitative discussion against the published tandem-mirror, direct-conversion, and data-center power-distribution literature (§6). Every headline quantity traces to a serial identifier in the Kronos Physics De-Risking Register [35], cited inline so each number is auditable.

2 Governing equations and the formal problem

2.1 *Fusion power and the neutron fraction*

The central cell burns deuterium and helium-3 through the primary aneutronic channel



with unavoidable deuterium–deuterium side reactions $\text{D} + \text{D} \rightarrow {}^3\text{He} + n(2.45\text{ MeV})$ and $\text{D} + \text{D} \rightarrow \text{T} + p$, the first of which, together with the D–T burn of the bred triton, supplies the residual neutron power. For a Maxwellian plasma the volumetric reaction rate between species j, k

is

$$\mathcal{R}_{jk} = \frac{n_j n_k}{1 + \delta_{jk}} \langle \sigma v \rangle_{jk}(T_i), \quad P_{\text{fus}} = \sum_{jk} \int_V \mathcal{R}_{jk} \mathcal{E}_{jk} dV, \quad (2)$$

with $\langle \sigma v \rangle$ the fusion reactivity and $\mathcal{E}_{jk}$ the reaction energy. The neutron fraction is the ratio of neutron-carried power to total,

$$f_n = \frac{P_n}{P_{\text{fus}}} = \frac{\int_V (\mathcal{R}_{\text{DDn}} E_n^{\text{DD}} + \mathcal{R}_{\text{DT}} E_n^{\text{DT}}) dV}{P_{\text{fus}}}, \quad (3)$$

and at the frozen operating point $P_{\text{fus}} = 4298.5$ MW, $P_n = 233.95$ MW, so $f_n = 5.44\%$ (register serial BU-L1-A1). The operating ion temperature $T_i \simeq 90$ keV sits just past the reactivity-weighted maximum of Q_E on the thermally stable branch, where $d(P_{\text{heat}} - P_{\text{loss}})/dT_i < 0$ so that the burn is self-correcting against a temperature excursion (BU-CX-30 line; a -29 MW keV $^{-1}$ restoring slope), an important property for a plant asked to hold a steady baseload behind a data center.

2.2 The plant energy multiplication

The quantity that decides whether the plant delivers net electricity is not the fusion gain but the plant energy multiplication—the ratio of gross electric output to the power that must be recirculated to sustain the plug and the thermal barrier. Writing P_{ch} for the axial charged-particle power that the DEC intercepts, P_{th} for the residual power taken through a thermal cycle at efficiency η_{th} , and $P_{\text{plug}}, P_{\text{aux}}$ for the recirculating plug and auxiliary powers,

$$Q_E = \frac{\eta_{\text{th}} P_{\text{th}} + \eta_{\text{DEC}} P_{\text{ch}}}{P_{\text{plug}} + P_{\text{aux}}}, \quad P_{\text{net}} = (P_{\text{plug}} + P_{\text{aux}})(Q_E - 1), \quad (4)$$

so that Q_E improves with recovered charged power but is bounded from below by the recirculating plug power. The frozen design point returns $Q_E = 1.318$ (BU-L1-A1). Because the recirculating denominator is set by the plug, the converter efficiency is a bounded upside lever and not the closure mechanism: we carry $\eta_{\text{DEC}} = 0.49$ as a deliberately conservative anchor—below the 0.56 engineering floor and the 0.70 Moir–Barr design value of the multi-stage-collector efficiency ladder, and well below the 0.85 ideal proton-only ceiling (BU-SX-01 adjudication; converter physics in [39])—so that no headline number leans on an optimistic conversion assumption.

2.3 The binding gate: the plug potential

Confining the central-cell ions against axial loss requires an electrostatic plug of potential ϕ_i , and the plug density needed to build it follows the Boltzmann relation between the confined and carrier populations,

$$\frac{n_p}{n_c} = \exp\left(\frac{e\phi_i}{T_e}\right), \quad (5)$$

so that at $\phi_i = 248.75$ keV and carrier temperature $T_e = 89.7$ keV the required ratio is $\exp(248.75/89.72) = 16$ (BU-L2-A5). Because the requirement sits inside the exponent, the carrier temperature is the strongest lever on it (figure 1); a 10% change in T_e moves the required ratio by roughly a

factor of 1.3. This is a requirement-class target: ϕ_i is more than two orders of magnitude beyond the demonstrated tandem-mirror plug-potential record set by GAMMA-10 [8, 7], and it is retired only by a dedicated plug-potential experiment at reactor-relevant scale, not by design study. We state this plainly and do not claim it retired; the potential is formed by an ECH-sustained hot-electron/sloshing-ion thermal barrier whose physics is a companion subject [41].

The confinement improvement the plug buys is the exponential enhancement of the ambipolar-well ion-confinement time over the collisional time. In the Pastukhov–Cohen theory of end-loss from an electrostatic mirror well [5, 6], the central-cell parallel confinement time scales as

$$\tau_{\parallel} \simeq \tau_{ii} \frac{2R_m}{\sqrt{\pi}(2R_m + 1) \ln(4R_m + 2)} \left(\frac{e\phi_i}{T_i}\right) \exp\left(\frac{e\phi_i}{T_i}\right), \quad (6)$$

with τ_{ii} the ion–ion collision time and R_m the plug mirror ratio; the exponential factor is the physical reason a modest-looking potential produces a large confinement gain, and the reason the required ϕ_i is so consequential. Equations (5)–(6) are the mathematical core of the burner: the entire net-power case rests on realizing $e\phi_i/T_e$, and Q_E closure is meaningful only conditional on it.

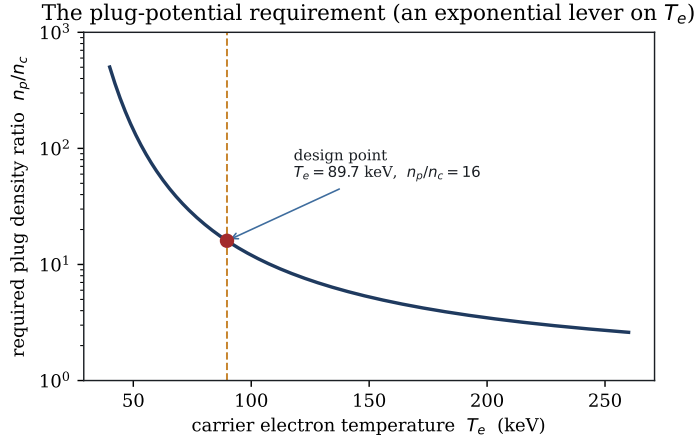


Figure 1. The plug-density requirement $n_p/n_c = \exp(e\phi_i/T_e)$ of equation (5) (BU-L2-A5). The design point sits at $n_p/n_c = 16$; the carrier temperature T_e is the exponential lever that sets it. This is the binding gate on which every net-power result is conditional.

2.4 Plug microstability

The plug's loss-cone distribution is a free-energy source for the drift-cyclotron loss-cone (DCLC) instability. Its marginal stability sits on the knee: a warm-ion fill fraction f_w inside the loss cone raises the growth-rate threshold, and the $\gamma = 0$ stabilization condition is met at

$$f_w \gtrsim f_c \approx \frac{n_{\text{warm}}}{n} \sim 6.25\% = \frac{1}{16}, \quad (7)$$

a *threshold* rather than a comfortable margin, and one that clears the DCLC mode only for sufficiently broad plug-edge density gradients (BU-L2-A5). Kinetic particle-in-cell modelling further finds that warm-fill quiets the DCLC branch but does not quench the anisotropy-driven Alfvén-ion-cyclotron (AIC) mode, which persists at $\gamma/\omega_{ci} \sim 0.037$ roughly independent of fill fraction (BU-L2-A7); the full microstability treatment, including the residual AIC activity and its consequences for the warm-fill budget, is a companion paper [43].

2.5 Synchrotron reabsorption and the length ladder

A distinctive result of the burner analysis is that the electron-cyclotron (synchrotron) loss is not a fixed fraction of fusion power but *scales with machine size*, which is what lets net output grow along a length ladder. Emission at harmonic n has an optical depth τ_n that falls with n ; the plasma is optically thick and re-emits as a black body up to a cutoff harmonic n^* where $\tau_{n^*} \sim 1$, and harmonics above n^* escape. The reabsorbed loss is set by the cutoff,

$$P_{\text{sync}} \simeq \int_{n^*}^{\infty} \eta(n) [1 - e^{-\tau_n}] dn, \quad n^* \propto \left(\frac{\omega_{pe}^2}{\omega_{ce}} a \right)^{1/3} \sim a^{1/3}, \quad (8)$$

so the cutoff grows as the one-third power of the minor radius a ; a larger n^* moves more emission into the reabsorbed band, so the synchrotron loss *fraction* falls as the machine grows. This is the analytic origin of the length ladder: net electric output of +104/ +850/ +2832 MW_e at 55/440/1400 m device length as fixed end losses are amortized over a longer central cell (KX-L1-A9), with the electron temperature pinned at the maximum- Q_E point by the synchrotron balance [42]. The MetroVolt reference campus is the mid-rung of this ladder, +850 MW_e at ~440–487 m.

2.6 Direct energy conversion

The charged-particle power that leaves the mirror ends is directed and monoenergetic-ish, which is what makes direct conversion possible. A multi-stage electrostatic (venetian-blind or travelling-wave) collector decelerates the ion stream against a graded potential and recovers its kinetic energy as terminal voltage; for a collector of N_s stages matched to a spread- Δ distribution the module efficiency approaches

$$\eta_{\text{DEC}} = 1 - \underbrace{\frac{\langle E - eV_{\text{coll}} \rangle}{\langle E \rangle}}_{\text{grid + spread + secondary losses}}, \quad \eta_{\text{DEC}} \xrightarrow{N_s \rightarrow \infty} \eta_{\text{ideal}}, \quad (9)$$

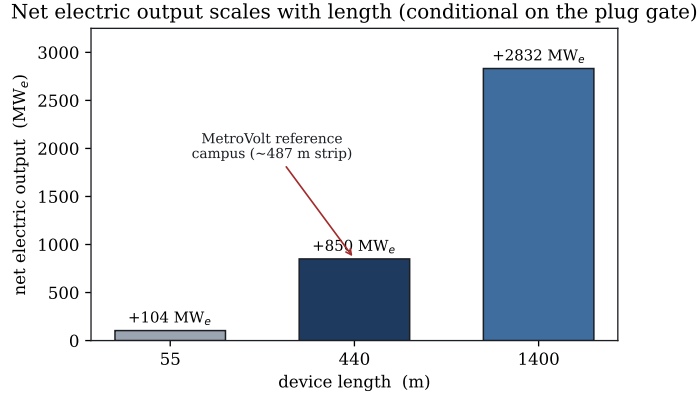


Figure 2. Net electric output versus device length (KX-L1-A9). Fixed end losses are amortized over a longer central cell; the MetroVolt reference campus is the mid-rung, +850 MW_e at ~440–487 m. Every value is conditional on retiring the plug-potential gate.

so that η_{DEC} rises with the number of graded stages and with the narrowness of the incident distribution. The register carries ~ 10 – 13 graded stages as sufficient for the 0.70 design value, and the efficiency is robust to plant throttling (falling only to 0.687 at 16 % load as ϕ_i shifts ~ 5 %; BU-CX-27). As stated above we anchor at the conservative $\eta_{\text{DEC}} = 0.49$. The physically important point for MetroVolt is not the exact efficiency but the *form* of the output: equation (9) produces its electricity as a terminal voltage of order $\phi_i \sim 249$ kV—high-voltage direct current—which is precisely what the data-center load consumes.

2.7 The data-center coupling: source-to-chip efficiency

The product physics is a power-electronics problem posed on the output of equation (9). A generator’s delivered efficiency to the silicon is the product of its conversion-stage efficiencies,

$$\eta_{\text{chain}} = \prod_{k=1}^{N_{\text{stage}}} \eta_k, \quad \eta_k \approx 1 - \varepsilon_k, \quad \varepsilon_k \gtrsim 0.6 \% \text{ per stage}, \quad (10)$$

so the count of stages, more than any single converter’s quality, sets the delivered efficiency. A thermal plant feeding an AC grid reaches the DC rack bus through rectification, UPS double conversion, transformation, and the rack power-supply unit—four-to-six stages—whereas a DEC delivering HVDC reaches the same bus through an HVDC→MVDC step and a rack-level DC–DC—two-to-three stages (BU-CX-01, BU-CX-02). The delivered figures are 75.9 % source-to-chip for the conventional AC path against 88.1 % for the DEC-direct medium-voltage-DC path, a +12.2-point gain (figure 3); energy-weighted over an annual load profile the DC-native chain holds 90.4 % against 78.9 % for AC because the DC chain does not sag at partial load the way a UPS does (BU-CX-06). This delivered-efficiency advantage is *independent of the plug gate*: it is a property of the

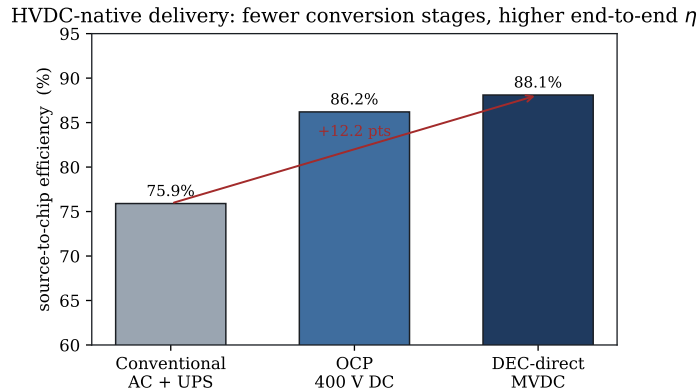


Figure 3. Source-to-chip electrical delivery efficiency (BU-CX-02, BU-CX-06). The DEC-direct MVDC path reaches the silicon in two-to-three conversion stages at 88.1 %, against 75.9 % for the conventional AC-plus-UPS path through four-to-six stages—a +12.2-point gain that is independent of the plug gate.

converter’s output form and the campus topology, not of the plasma closure.

3 Computational methodology (GPU-accelerated HPC)

Every gate above was reduced on the NVIDIA GPU HPC campaign, with the reduced-order and control-loop layers on the in-house digital-twin node. We name the codes so each result is reproducible.

Zero-dimensional power balance. The design point ($Q_E, P_{\text{fus}}, P_n, f_n$) and the plug requirement of equations (4)–(5) were solved with the Kronos reduced-order mirror solver (`solve_mirror` in the open Kronos toolkit), a Pastukhov-plug 0-D balance that returns $Q_E = 1.318$, $P_{\text{fus}} = 4298.5$ MW, $P_n = 233.95$ MW bit-for-bit (BU-L1-A1). A design-space sweep over density and temperature (BU-L2-A2) returned $Q_E > 1$ across the sampled box (25/25 points, 1.107–1.414) with a reactivity ridge at $T_i \simeq 85$ keV.

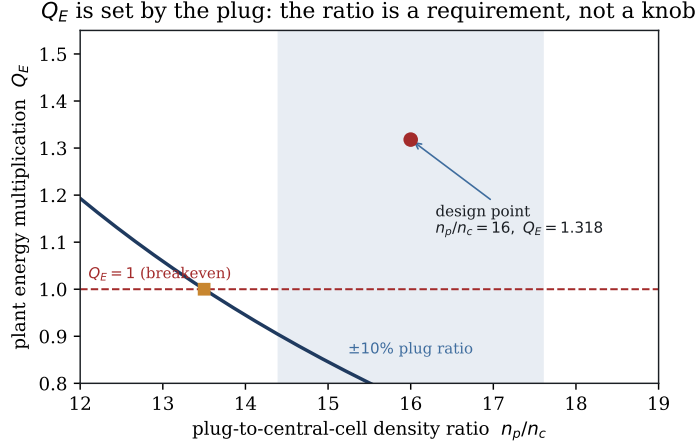


Figure 4. Plant energy multiplication versus the plug-to-central-cell density ratio (BU-CX-26). The design point sits at $n_p/n_c = 16$, $Q_E = 1.318$; breakeven is at $n_p/n_c \approx 13.5$, so the net-power margin $Q_E - 1$ changes by roughly half over a $\pm 10\%$ excursion in the plug ratio—the ratio is a requirement, not a free knob. The curve is a two-parameter fit through the frozen breakeven and design-point anchors.

Kinetic confinement and microstability. The loss-cone distribution and its stability were treated with Fokker–Planck and particle-in-cell kinetics. The DCLC warm-fill threshold of equation (7) and the residual AIC branch were computed with the electromagnetic PIC codes WarpX and Smilei (BU-L2-A5, BU-L2-A7); the strong-plug Fokker–Planck balance and RF alpha-channeling estimate used a reduced two-dimensional velocity-space Fokker–Planck model with an ICRF ray-tracing driver (GENRAY/CQL3D-class), which lifts the recoverable charged-ion fraction toward 80 % under localized-plug assumptions (BU-L2-A8).

Uncertainty quantification. The Q_E distribution of figure 5 is a 4×10^4 -sample Monte-Carlo propagation over the design input ranges, returning $Q_E = 1.317 \pm 0.14$ with $P(Q_E > 1) = 1.0$, the central-cell β the dominant driver (KX-L1-A5-BU). The plug-ratio sensitivity of figure 4 is a

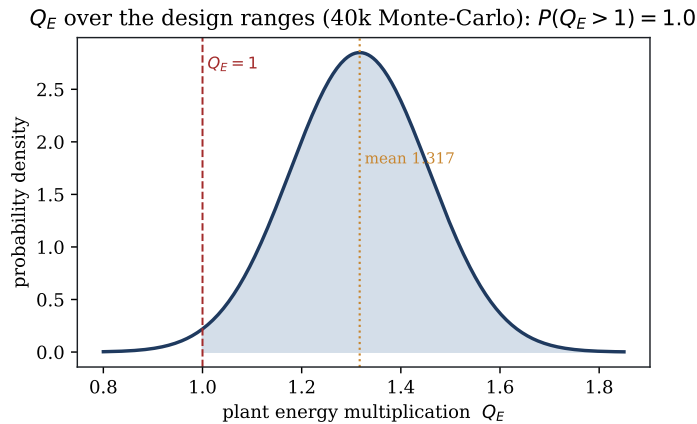


Figure 5. Plant-energy-multiplication uncertainty over the design ranges (KX-L1-A5-BU; 4×10^4 Monte-Carlo samples): $Q_E = 1.317 \pm 0.14$ with $P(Q_E > 1) = 1.0$, the central-cell β the dominant driver. The design closes robustly across its input ranges, conditional on the plug gate.

`solve_mirror` scan (BU-CX-26).

Neutronics and siting. The urban dose and shield thickness were estimated by point-kernel attenuation with an OpenMC cross-check pathway (BU-CX-28), and the mobile tritium source term by an in-vessel inventory balance (BU-CX-29).

Power electronics and thermal integration. The DC delivery chain, the fault-ride-through, the electronic load-following, and the waste-heat-to-cooling integration were modelled with circuit-level electromagnetic-transient and system-efficiency solvers (BU-CX-01 through BU-CX-16); the closed-loop bus-regulation study of figure 11 is a control simulation holding the DC bus to $\pm 0.40\%$ through stepped load transients at constant plasma baseload (BU-CX-07). No result in this section reports, or depends on, any cost, price, or financial quantity; the computational campaign is reported in engineering units only.

4 The direct-DC data-center burner

4.1 HVDC-native delivery and fewer conversion stages

The MetroVolt architecture (figure 6) delivers the DEC's high-voltage DC output through a

MetroVolt: behind-the-meter, HVDC-native, islandable data-center burner

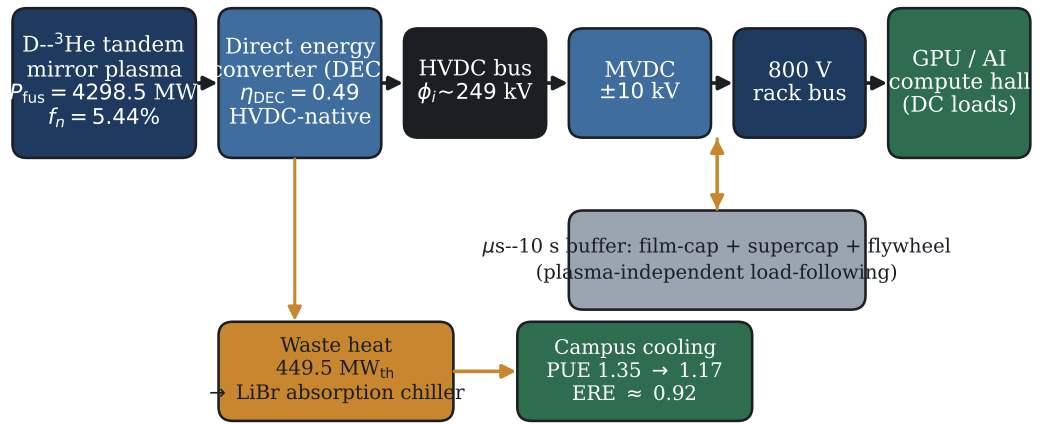


Figure 6. (Schematic.) The MetroVolt direct-DC campus architecture. The DEC delivers high-voltage direct current natively; an MVDC campus bus feeds an 800 V rack bus and the GPU hall through two-to-three conversion stages, with a plasma-independent electronic buffer on the DC bus and waste-heat-driven absorption cooling on the thermal branch. No data values are plotted; the figure encodes topology only.

medium-voltage DC (MVDC, ± 10 kV) campus bus to an 800 V rack bus, deleting the plant AC generator, the grid step-up, the UPS double conversion, and the rack AC power-supply front end. The topology optimum is a two-stage HVDC $\rightarrow$ MVDC $\rightarrow$ 800 V path at $\sim 98.2\%$ conversion, and the MVDC campus bus cuts distribution copper by a large factor relative to an 800 V-only campus at equal delivered power because I^2R loss falls as the square of the raised bus voltage (BU-CX-01). No solid-state transformer is required: the DEC is HVDC-native, so the single isolation stage collapses into one silicon-carbide dual-active-bridge optimized near 15.8 kHz at 98.85%, and the DEC-direct path beats an SST-fed path by +3.0 points with one fewer stage (BU-CX-04). The delivered source-to-chip efficiencies of $\S 2$ —88.1% DC-native versus 75.9% AC—are the headline product result (figure 3).

The DC bus is also inherently better-behaved for an IT load. The DEC output is smooth: 1% ripple needs only a $\sim 234 \mu\text{F}$ bus capacitance, far below the millifarad-scale hold-up that the ITIC ride-through curve already demands, so no active harmonic filtering is required (BU-CX-03). And the DEC's intrinsic current limit caps a bus fault at ~ 3 kA against ~ 50 kA from a stiff grid; a $2 \mu\text{s}$ SiC solid-state circuit breaker bounds a bolted MVDC short to ~ 4 kA, 1.3% of the undamped LC peak (BU-CX-05)—a safety property, not merely an efficiency one.

4.2 Electronic load-following at constant plasma baseload

A fusion plasma cannot follow a data center's transients: its throttle authority is of order 0.76 MW s^{-1} , while synchronized GPU training swings ramp at $\sim 80 \text{ MW s}^{-1}$ (95th percentile), about $105\times$ faster (BU-CX-08). MetroVolt therefore does not ask the plasma to follow the load. It runs the reactor at a constant baseload and interposes a plasma-independent electronic buffer—a film-capacitor/supercapacitor/flywheel hybrid of order a few tonnes and $\sim 8.4 \text{ MJ}$ store per 30 MW

hall—that absorbs the transients on microsecond-to-second timescales while the DC bus is held within $\pm 0.40\%$ through $60 \rightarrow 92 \rightarrow 40 \rightarrow 75\%$ load steps (figure 11; BU-CX-07, BU-CX-09). Because the store buffers transients rather than bulk energy, no lithium-ion energy reservoir is needed; a baseload plant with a small fast buffer is the natural firm supply for a crest-factor load. The same DC-native buffer makes the plant grid-forming: a sub-10 ms fast-frequency response lifts a 6%-loss frequency nadir by $+0.53$ Hz, above the under-frequency load-shed line, and supports black-start and islanded operation that a grid-following plant cannot (BU-CX-10, BU-CX-11).

4.3 Waste heat, cooling, and power-usage effectiveness

The residual (non-direct) power leaves the plant as heat, and a data center is a large heat sink with an even larger cooling load, so the two integrate naturally. At the 55 m rung the plant rejects $449.5 \text{ MW}_{\text{th}}$ of waste heat, of which $\sim 142.9 \text{ MW}_{\text{th}}$ drives a double-effect lithium-bromide absorption chiller ($\text{COP} \approx 1.35$) that covers the full cooling demand of a 100-MW-IT campus on $\sim 74 \text{ MW}_{\text{th}}$, cutting the electric cooling load by $\sim 91\%$ (from 16.7 to 1.5 MW_e ; BU-CX-12, BU-CX-08). Combined with DC-native conversion and direct-liquid cooling that captures $\sim 90\%$ of the IT power at 55°C for reuse (BU-CX-13), the facility power-usage effectiveness falls from a baseline PUE of 1.35 to 1.17 (figure 7): -0.05 from DC-native conversion, -0.07 from direct-liquid cooling, -0.04 from waste-heat absorption cooling, and -0.02 from auxiliaries (BU-CX-14). With even a quarter of the waste heat taken up off-site the energy-reuse effectiveness reaches ERE ≈ 0.92 —the campus exports more useful energy than its own overhead consumes.

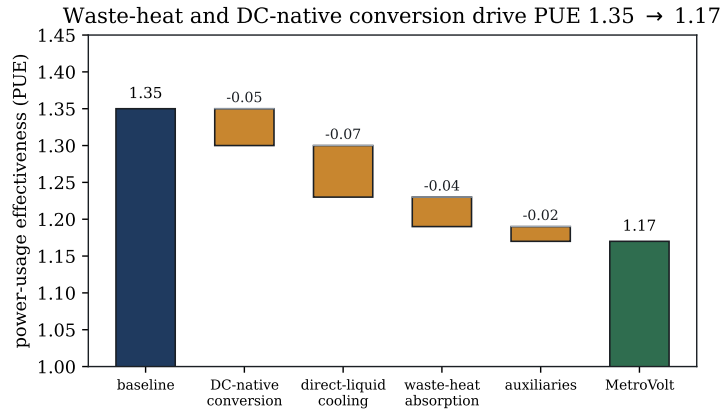


Figure 7. Power-usage-effectiveness pathway (BU-CX-14). DC-native conversion, direct-liquid cooling, waste-heat-driven absorption cooling, and auxiliary savings take the facility from a baseline PUE of 1.35 to 1.17 ; with off-site heat uptake the energy-reuse effectiveness reaches ERE ≈ 0.92 .

4.4 Power density and behind-the-meter siting

A compact mirror is a linear device, and co-locating generation with compute along a right-of-way gives a site power density of $\sim 22 \text{ MW}$ per acre (850 MW on ~ 39 acres), roughly $3\times$ a gas-plus-air-cooled data center, $\sim 99\times$ a solar-backed campus, and $\sim 700\times$ a wind-backed one (figure 8; BU-CX-17, MV-L1-A7). Behind-the-meter operation raises delivered efficiency by a further 6.3 points (avoiding transmission and distribution loss, 98.5 versus 92.2%) and, more consequentially for deployment, skips the interconnection queue: time-to-power falls from ~ 78 to ~ 30 months (BU-CX-18). A modest grid tie is retained for export and ancillary services, so the optimal topology is a behind-the-meter hybrid rather than a fully islanded plant.

4.5 The capacity-factor case

The reason a firm baseload source is the right generator for a 24/7 compute load is a capacity-factor argument. To firm a continuous load L with a source of capacity factor CF requires nameplate capacity $C \gtrsim L/\text{CF}$ plus storage to bridge the intermittency,

$$C \gtrsim \frac{L}{\text{CF}}, \quad S \gtrsim L \Delta t_{\text{lull}}, \quad (11)$$

with Δt_{lull} the worst-case low-output interval. Baseload fusion at $\text{CF} \approx 0.93$ needs $\sim 1.2\times$ the load with buffer-only firming; solar and wind at $\text{CF} = 0.24\text{--}0.35$ need $\sim 3.3\text{--}4.8\times$ the capacity plus $\sim 100\text{--}215$ hours of storage to reach the same reliability (figure 9; BU-CX-23). For a campus whose

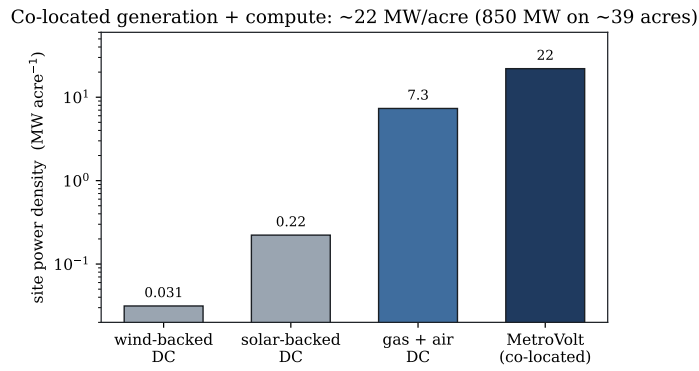


Figure 8. Site power density (BU-CX-17). Co-located generation and compute reach ~ 22 MW per acre (850 MW on ~ 39 acres), roughly $3\times$ a gas-plus-air-cooled data center and orders of magnitude above renewable-backed campuses. Axis is logarithmic.

economics are set by continuous utilization, the oversizing and storage burden of an intermittent source is the decisive engineering difference, independent of any cost figure.

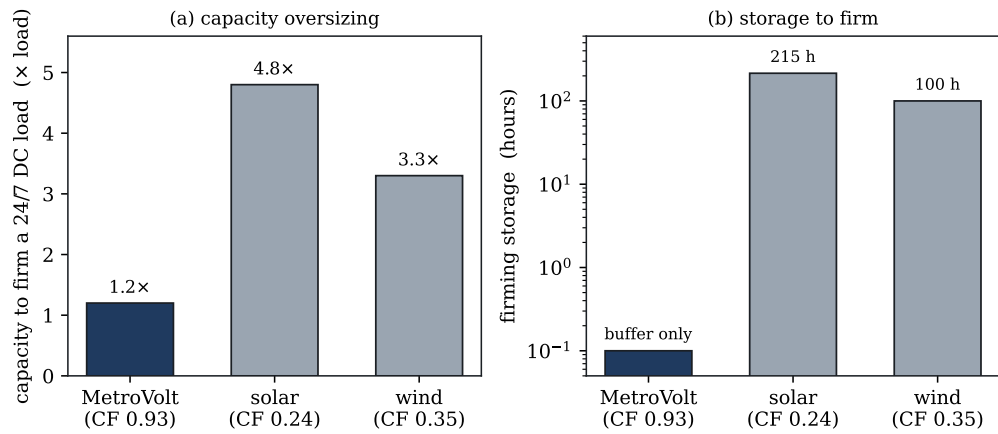


Figure 9. The capacity-factor case for a firm baseload source (BU-CX-23). **(a)** Firming a 24/7 load needs $\sim 1.2\times$ nameplate from baseload fusion (CF 0.93) but $3.3\text{--}4.8\times$ from solar or wind. **(b)** The renewable paths additionally need 100–215 hours of firming storage; the fusion path needs only the transient buffer.

4.6 Neutron-lean urban siting and the tritium source term

The low-neutron burn is what makes an urban or near-campus site feasible. A point-kernel dose estimate meets an urban public-dose target of $\sim 0.11 \mu\text{Sv h}^{-1}$ at 50 m behind a buildable ~ 2.8 m concrete biological shield, against ~ 3.06 m for a deuterium–tritium plant of equal power—a modest saving, because dose scales logarithmically with shield thickness (BU-CX-28). The licensing-decisive quantity is not the shield but the mobile tritium source term. Although the burner self-generates ~ 32 kg of tritium per full-power year (of which $\sim 31\%$ is burned in situ), the licensing-relevant mobile inventory—held to a one-hour processing buffer—is only ~ 3.6 g, roughly $275\times$ below the ~ 1 kg-class inventory of a D–T plant (figure 10; BU-CX-29). This small source term is what underwrites siting a fusion plant next to a data-center campus and is the physical basis of the low-activation licensing case [45].

5 Results and verification

Table 1 collects the frozen design point; figures 6–10 present the physics and product results in turn. Table 2 collects the data-center product metrics with their register serials and their dependence—or independence—from the plug gate.

6 Discussion

6.1 Against the tandem-mirror record

The physics honesty of this design rests on separating what is demonstrated from what is required. The tandem-mirror concept and its thermal-barrier variant are mature in the literature—Fowler

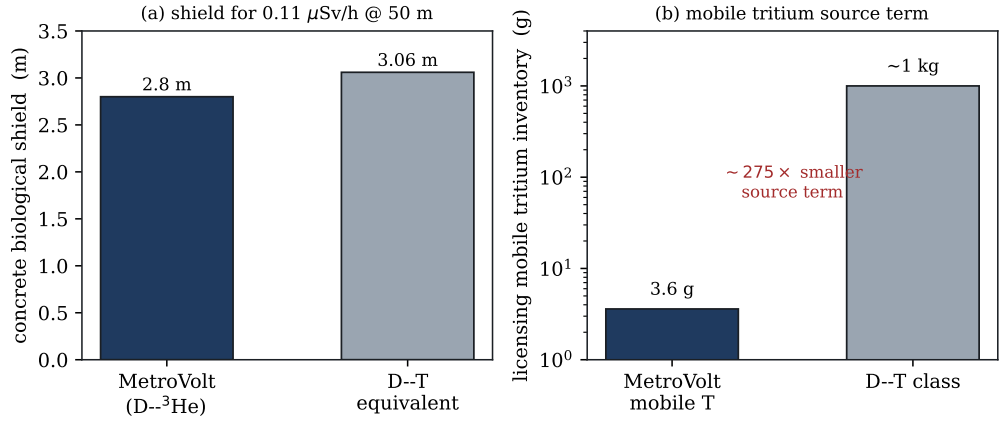


Figure 10. Neutron-lean urban siting (BU-CX-28, BU-CX-29). **(a)** The low-neutron burn meets an urban public-dose target behind a ~ 2.8 m concrete shield versus ~ 3.06 m for a D–T equivalent. **(b)** The licensing-relevant mobile tritium inventory is ~ 3.6 g, roughly $275\times$ below a D–T-class plant; axis is logarithmic.

Table 1. MetroVolt burner design point (register serials BU-L1-A1 / KX-L1-A9 / BU-L2-A5 / KX-L1-A5-BU). Every net-power quantity is conditional on retiring the plug-potential gate; the neutron-source performance is not.

quantity	value
plant energy multiplication Q_E	1.318
fusion power P_{fus}	4298.5 MW
neutron power P_n	233.95 MW
neutron fraction f_n	5.44 %
plug potential ϕ_i	248.75 kV
carrier temperature T_e	89.7 keV
plug density ratio n_p/n_c	16
converter efficiency η_{DEC} (conservative anchor)	0.49
net electric (55/440/1400 m)	+104/ + 850/ + 2832 MW _e
$P(Q_E > 1)$ over design ranges	1.0

Table 2. MetroVolt data-center product metrics, their register serials, and their dependence on the plug gate. “Robust” quantities hold regardless of the plug closure; “conditional” quantities scale with the net-power case.

metric	value	serial	plug dependence
source-to-chip efficiency (DC-native / AC)	88.1 % / 75.9 %	BU-CX-02	robust
energy-weighted annual η (DC / AC)	90.4 % / 78.9 %	BU-CX-06	robust
load-following ramp (buffer / plasma)	80 / 0.76 MW s ^{−1}	BU-CX-08	robust
DC-bus regulation across load steps	± 0.40 %	BU-CX-07	robust
power-usage effectiveness	1.35 $\rightarrow$ 1.17	BU-CX-14	robust
site power density	~ 22 MW acre ^{−1}	BU-CX-17	conditional
firming capacity vs. renewables	1.2 $\times$ vs. 3.3–4.8 $\times$	BU-CX-23	robust
urban shield / D–T equivalent	2.8 / 3.06 m	BU-CX-28	robust
mobile tritium source term	~ 3.6 g	BU-CX-29	robust
net electric (reference campus)	+850 MW _e	KX-L1-A9	conditional

and Logan’s tandem-mirror reactor [1], the axisymmetrized GAMMA-10 experiments that formed thermal barriers and confining potentials [7, 8], and the Pastukhov/Cohen theory of ambipolar-well end-loss [5, 6] underlying equation (6). What no experiment has demonstrated is a plug potential at the reactor-relevant magnitude of equation (5): the required ϕ_i is more than two orders of magnitude beyond the GAMMA-10 record, and the WHAM-class high-field mirror now coming online addresses the unplugged confinement regime rather than the plugged branch our net-power case depends on (BU-L2-A3). We therefore present the plug requirement as the binding open gate, retired only by a dedicated plug-potential experiment, and every net-power figure in table 1 and figure 2 inherits that condition. The neutron-source performance— $f_n = 5.44$ %, $P_n = 233.95$ MW—is by contrast independent of the plug and rests on the fusion cross-sections alone.

6.2 Against the direct-conversion literature

Direct energy conversion of a mirror’s escaping ion stream has a long pedigree, from the Post venetian-blind collector through the Moir–Barr analyses that set the multi-stage efficiency expectations near 0.70 [9, 10, 11], and modern reappraisals of expander-region conversion [12]. Our

conservative anchor of $\eta_{\text{DEC}} = 0.49$ sits deliberately below that literature’s design value, so no headline number leans on an optimistic converter; the efficiency ladder (0.49 floor anchor, 0.70 design, 0.85 ideal) is carried explicitly (BU-SX-01). The contribution specific to MetroVolt is not a new converter efficiency but the observation that the converter’s *output form*—high-voltage DC at the plug scale, equation (9)—is a first-class product advantage for a data-center load, and the quantification of that advantage as a +12.2-point source-to-chip efficiency gain (figure 3). A companion paper develops the DEC-to-load power-electronics interface in full [40].

6.3 Against the data-center power-distribution literature

The data-center community has independently converged on DC distribution and direct-liquid cooling as the levers on facility efficiency: the Open Compute Project’s 48-to-400 V DC racks, the migration to direct-liquid and two-phase immersion cooling, and the PUE and energy-reuse-effectiveness metrics standardized for reporting facility overhead [22, 23]. MetroVolt’s advantage is that it supplies the front of that chain in the native form the chain wants: a generator whose output is already HVDC removes the rectification and UPS double conversion that a grid-fed facility cannot, which is why the DC-native chain holds 90.4 % against 78.9 % energy-weighted (BU-CX-06). The load-following result (figure 11) similarly reframes a perceived fusion liability—slow plasma response—as a non-issue: a baseload plant with a small electronic buffer follows a crest-factor AI load better than a dispatchable thermal plant chasing it with steam, because the buffer responds in milliseconds and the plasma never moves. The capacity-factor argument of equation (11) and

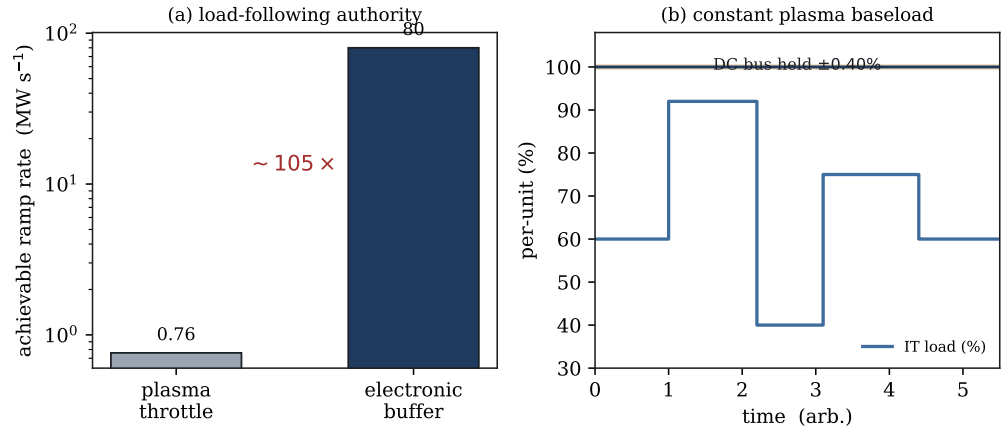


Figure 11. Electronic load-following at constant plasma baseload (BU-CX-07, BU-CX-08). **(a)** The electronic buffer follows load at $\sim 80 \text{ MW s}^{-1}$, about $105\times$ the plasma throttle authority. **(b)** Across $60 \rightarrow 92 \rightarrow 40 \rightarrow 75\%$ IT-load steps the closed-loop controller holds the DC bus within $\pm 0.40\%$ while the plasma runs at a constant baseload; the trace is illustrative, the band and load steps are the frozen result.

figure 9 is the quantitative statement of why a firm source, not an intermittent one oversized and storage-backed, is the right generator for a continuous compute load.

6.4 Two distinct magnets

Throughout, the burner plug coil and the sister breeder centrepost are treated as the distinct devices they are. The breeder centrepost of the companion Hyperion platform [36] governs at a peak field of 16.84 T; the MetroVolt burner plug coil is a separate, higher-field REBCO magnet at 26.49 T, structurally qualified by analytic thick-wall and finite-element analysis (magnetic pressures 115/279/358 MPa at 17/26.49/30 T against a 667 MPa allowable, FEA stress ratio PASS; BU-L1-A2) and detailed in the magnet-basis companion [38]. Conflating the two fields is a category error the design explicitly avoids.

7 Limitations

Four caveats bound the claims, each stated with its evidence class. First, every net-power number— Q_E , the length ladder, the reference-campus $+850 \text{ MW}_e$ —is contingent on retiring the plug-potential gate of equation (5), which a design study cannot do. Second, the converter efficiency is a declared, deliberately conservative input, not a computed plant result; the direct-DC *topology* advantage, however, does not depend on its value. Third, the plug microstability is marginal: warm-fill clears the DCLC knee only as a threshold (equation (7)) and a residual AIC

branch persists (BU-L2-A7). Fourth, the helium-3 fuel supply is a separate, lunar-gated problem treated elsewhere [44] and outside this power-plant design. The data-center product metrics of table 2—the delivery efficiency, load-following, PUE, and siting case—are the robust, plug-independent core; the net-power magnitude is the conditional part.

8 Conclusion

MetroVolt matches the shape of the modern data-center load because its generator produces electricity in the form the load consumes. The D-³He tandem-mirror burner reaches a plant energy multiplication $Q_E = 1.318$ at 4298.5 MW fusion power with a neutron fraction of only 5.44 %, delivering +104/ + 850/ + 2832 MW_e across a length ladder with $P(Q_E > 1) = 1.0$; its direct energy converter produces high-voltage DC natively, reaching the silicon at 88.1 % source-to-chip against 75.9 % for a conventional AC plant, following a crest-factor AI load electronically at constant plasma baseload, taking the facility to PUE 1.17 through waste-heat integration, and licensing against a ~3.6 g mobile tritium source term small enough to site beside a campus. The binding physics gate is the plug-potential requirement $n_p/n_c = 16$, retired only by experiment; the burner plug coil (26.49 T) and breeder centrepost (16.84 T) are modelled as the distinct magnets they are. The design is presented with its robust and conditional parts separated openly—the product coupling robust, the net-power magnitude conditional on the one residual physics gate.

Acknowledgments

Part of the Kronos Fusion Energy 2026 design series. The authors thank the Kronos physics de-risking programme for the frozen result set underlying every quantitative claim, and the reviewers of the 2026 design series.

Reproducibility. Every headline quantity is a frozen anchor in the Kronos Physics De-Risking Register [35] (doi:10.5281/zenodo.22133057): the burner power balance and neutron fraction (BU-L1-A1), the plug-potential requirement (BU-L2-A5), the Q_E uncertainty (KX-L1-A5-BU), the length ladder (KX-L1-A9), and the MetroVolt product metrics (BU-CX-01 through BU-CX-29, MV-L1-A1 through MV-L1-A9). Each is regenerable from the open, CC-BY-4.0 deposit; this paper is archived at doi:10.5281/zenodo.22645695; official version: <https://www.kronosfusionenergy.com/publications/metrovolt-burner.html>. No result depends on unpublished or proprietary data.

Scope — what this paper does not claim. Q_E is the engineering gain at the frozen design point, not a demonstrated result; the end-plug density requirement is stated as the binding open gate, not solved; “low-neutron” is not “aneutronic”; and no cost, price, tariff, valuation, or other financial quantity is reported or implied anywhere in this paper.

Data availability

The figure-generation script and the frozen anchors used for figures 6–10 are archived with the deposit at doi:10.5281/zenodo.22645695, which cross-references the Kronos 2026 series: the register [35], the Hyperion breeder platform [36], the sister Aegis burner [37], the REBCO magnet basis [38], direct energy conversion [39], the DEC-to-load data-center interface [40], recirculation-resolved Q_E closure [41], synchrotron transport [42], velocity-space microstability [43], the helium-3 supply gate [44], and the Part 30 licensing thesis [45]. Any economic analysis is reported separately and is not part of the public deposit.

Competing interests

Provisional patent applications relating to aspects of this work have been filed.

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